Most Asked SQL Interview Questions

Fresher / Placement Interview Preparation

1. Find second highest salary

Answer:

```
SELECT DISTINCT salary
FROM employees
ORDER BY salary DESC
LIMIT 1 OFFSET 1;
```

2. Find nth highest salary

Answer:

```
SELECT *
FROM (
SELECT e.*,
DENSE_RANK() OVER (ORDER BY salary DESC) AS rn
FROM employees e
) x
WHERE rn = N;
```

3. Find duplicate records

Answer:

```
SELECT name, COUNT(*) AS cnt
FROM employees
GROUP BY name
HAVING COUNT(*) > 1;
```

4. Delete duplicate records

Answer:

```
WITH CTE AS (
SELECT *,
ROW_NUMBER() OVER (
PARTITION BY name ORDER BY id
) AS rn
FROM employees
)
DELETE FROM CTE
WHERE rn > 1;
```

5. Find highest salary in each department

Answer:

```
SELECT department, MAX(salary) AS max_salary
FROM employees
GROUP BY department;
```

6. Find top 3 salaries

Answer:

```
SELECT *
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

7. Find employees earning above average salary

Answer:

```
SELECT *
FROM employees
WHERE salary > (SELECT AVG(salary) FROM employees);
```

8. Find employees without a department

Answer:

```
SELECT *
FROM employees
WHERE department IS NULL;
```

9. Find customers who never placed an order

Answer:

```
SELECT *
FROM customers c
WHERE NOT EXISTS (
SELECT 1
FROM orders o
WHERE o.customer_id = c.customer_id
);
```

10. Find number of employees in each department

Answer:

```
SELECT department, COUNT(*) AS employee_count
FROM employees
GROUP BY department;
```

11. Find department with highest average salary

Answer:

```
SELECT department, AVG(salary) AS avg_salary
FROM employees
GROUP BY department
ORDER BY avg_salary DESC
LIMIT 1;
```

12. Find duplicate/consecutive records

Answer:

```
-- Duplicate values
SELECT name, COUNT(*)
FROM employees
GROUP BY name
HAVING COUNT(*) > 1;

-- Consecutive IDs
SELECT e1.id
FROM employees e1
JOIN employees e2 ON e2.id = e1.id + 1;
```

13. Difference between highest and lowest salary

Answer:

```
SELECT MAX(salary) - MIN(salary) AS salary_difference
FROM employees;
```

14. Employees whose salary is greater than their manager's

Answer:

```
SELECT
e.name AS employee,
e.salary AS employee_salary,
m.name AS manager,
m.salary AS manager_salary
FROM employees e
JOIN employees m
ON e.manager_id = m.id
WHERE e.salary > m.salary;
```

15. Find latest order of each customer

Answer:

```
SELECT *
FROM (
SELECT o.*,
ROW_NUMBER() OVER (
PARTITION BY customer_id
ORDER BY order_date DESC
) AS rn
FROM orders o
) x
WHERE rn = 1;
```

16. Find employees with the same salary

Answer:

```
SELECT salary, COUNT(*)
FROM employees
GROUP BY salary
HAVING COUNT(*) > 1;
```

17. Find highest salary employee

Answer:

```
SELECT *
FROM employees
WHERE salary = (SELECT MAX(salary) FROM employees);
```

18. Find highest salary employee(s) in each department

Answer:

```
SELECT *
FROM (
SELECT e.*,
RANK() OVER (
PARTITION BY department
ORDER BY salary DESC
) AS rn
FROM employees e
) x
WHERE rn = 1;
```

19. Find second highest salary in each department

Answer:

```
SELECT *
FROM (
SELECT e.*,
DENSE_RANK() OVER (
```

```
PARTITION BY department
ORDER BY salary DESC
) AS rn
FROM employees e
) x
WHERE rn = 2;
```

20. Find departments having more than 5 employees

Answer:

```
SELECT department, COUNT(*) AS employee_count
FROM employees
GROUP BY department
HAVING COUNT(*) > 5;
```

21. Find employees whose name starts with A

Answer:

```
SELECT *
FROM employees
WHERE name LIKE 'A%';
```

22. Find employees whose name ends with n

Answer:

```
SELECT *
FROM employees
WHERE name LIKE '%n';
```

23. Find employees whose name contains 'an'

Answer:

```
SELECT *
FROM employees
WHERE name LIKE '%an%';
```

24. Count total employees

Answer:

```
SELECT COUNT(*)
FROM employees;
```

25. Find average salary

Answer:

```
SELECT AVG(salary)
FROM employees;
```

26. Find employees hired in a particular year (MySQL)

Answer:

```
SELECT *
FROM employees
WHERE YEAR(hire_date) = 2025;
```

27. Find departments with no employees

Answer:

```
SELECT d.department_name
FROM department d
```

```
LEFT JOIN employees e
ON d.department_id = e.department_id
WHERE e.id IS NULL;
```

28. Find customers who placed more than 3 orders

Answer:

```
SELECT customer_id, COUNT(*) AS order_count
FROM orders
GROUP BY customer_id
HAVING COUNT(*) > 3;
```

29. Find total order amount for each customer

Answer:

```
SELECT customer_id, SUM(amount) AS total_amount
FROM orders
GROUP BY customer_id;
```

30. Find customer who spent the most

Answer:

```
SELECT customer_id, SUM(amount) AS total_spent
FROM orders
GROUP BY customer_id
ORDER BY total_spent DESC
LIMIT 1;
```

SQL Theory Questions to Prepare

1. Primary key vs Foreign key
2. UNIQUE key vs PRIMARY KEY
3. INNER JOIN vs LEFT JOIN vs RIGHT JOIN
4. WHERE vs HAVING
5. DELETE vs TRUNCATE vs DROP
6. UNION vs UNION ALL
7. Subquery vs correlated subquery
8. IN vs EXISTS vs NOT IN vs NOT EXISTS
9. IS NULL vs = NULL
10. Aggregate functions vs scalar functions
11. RANK() vs DENSE_RANK() vs ROW_NUMBER()
12. What is a CTE (Common Table Expression)?
13. What is a SELF JOIN?
14. View vs Stored Procedure
15. What is an Index?
16. Normalization: 1NF, 2NF, 3NF, BCNF
17. Transactions and ACID properties
18. COMMIT, ROLLBACK, SAVEPOINT
19. LEAD() and LAG()
20. Running total and latest record per group

Priority for placement interviews: SELECT/WHERE/ORDER BY → GROUP BY/HAVING → JOINS → Subqueries → IN/EXISTS → Aggregate functions → nth highest salary → duplicates → window functions → UNION → CTE/SELF JOIN → keys/normalization/indexes/transactions.